SWA-DPBR: Socially-Weighted Alignment with Dual-Pass Bootstrap Reliability for Cross-Cultural Moral Alignment of Large Language Models

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Abstract

Large language models make moral judgments as if the whole world thinks like the West—yet 40 million human responses across 233 countries say otherwise. We show that a frozen LLM can be taught to *respect* that diversity *without any training*. We introduce **SWA-DPBR** (Socially-Weighted Alignment with Dual-Pass Bootstrap Reliability), a training-free, inference-time method that steers binary moral choices toward country-specific human preferences using only logit corrections—no weight updates, no per-country finetuning, no white-box access. The key insight is that *disagreement is signal*: we instantiate each country as a panel of World Values Survey–grounded persona agents whose within-country disagreement pinpoints exactly where and how much to correct the model's default. That structured disagreement is converted into a bounded logit shift via Prospect-Theory importance sampling, stabilised through a dual-pass bootstrap reliability gate that automatically shrinks corrections when sampling is noisy. Evaluated on MultiTP across **twenty countries, four continents**, and six open model families (3.8B–70B parameters), SWA-DPBR reduces misalignment by **19–24%** on the strongest checkpoints, with gains spanning the Americas, Europe, and Asia—not confined to any single region. When the method fails, the culprit is collapsed decision-logit entropy, an architectural bottleneck rather than a cultural blind spot, suggesting that inference-time cultural steering is broadly viable wherever the base model maintains a well-behaved decision surface. Code: [repository].

1 Introduction

A self-driving car must choose: spare a group of elderly men or a group of young women. Ask in Tokyo and Bogotá and you will get systematically different answers. The Moral Machine experiment [Awad et al., 2018] demonstrated this at scale: 40 million judgments from 233 countries revealed that moral preferences are not noise—they are *structured* by culture. East Asian respondents weight youth more heavily; Southern-hemisphere populations show distinct utilitarian calculi; Islamic societies prioritise different social roles. Yet the large language models increasingly enlisted to mediate such decisions encode a single, culturally monolithic moral stance inherited from predominantly Western training data [Henrich et al., 2010].

The scale of the problem. The multilingual MultiTP benchmark [Jin et al., 2025] now quantifies the answer: country-specific Average Marginal Component Effects (AMCE) across six moral dimensions and 107 languages. The verdict is damning: **none** of 19 contemporary LLMs achieves broadly satisfactory alignment; most exhibit WEIRD-biased reasoning that departs sharply from non-WEIRD human preferences.

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

Fixing it without retraining. A practical cross-cultural decoder must satisfy three constraints simultaneously: **(C1)** no weight updates, **(C2)** no per-culture supervised labels at scale, and **(C3)** no white-box access to internal activations. Every existing approach trades at least one away: fine-tuning [Bai et al., 2022, Ouyang et al., 2022] violates C1 and C2; prompting-based steering lacks empirical grounding [Jin et al., 2025]; activation steering [Arditi et al., 2025, Zou et al., 2023] requires C3; reward-guided decoding [Khanov et al., 2024, Mudgal et al., 2024] assumes per-country reward models that do not exist; and culture-aware adapters [Li et al., 2024, Yao et al., 2024] need per-culture training, scaling poorly past a handful of countries.

We propose **SWA-DPBR** (Socially-Weighted Alignment with Dual-Pass Bootstrap Reliability), a method that satisfies all three constraints by exploiting a simple but powerful insight: *within-country disagreement is the steering signal itself.* Moral preferences within a society are not monolithic—age cohorts and social strata prioritise fundamentally different trade-offs [Awad et al., 2018, Kleiman-Weiner et al., 2017]. The World Values Survey [Haerpfer et al., 2022] provides empirical structure for that variation. SWA-DPBR instantiates each country as a small panel of culturally grounded persona agents drawn from WVS microdata; when those agents *disagree* on a dilemma, the spread of their logit gaps *pinpoints* where a correction is needed and how large it should be. When they agree, the vanilla model is typically already close—no correction is applied.

The operational pipeline is a single additional batched forward pass: WVS-grounded personas + base model, native-language prompting, positional debiasing, then two independent Prospect-Theory importance-sampling passes [Kahneman and Tversky, 1979] whose outputs are merged with a bootstrap reliability weight—if the two passes disagree, the correction automatically shrinks toward zero. An effective-sample-size guard and a light hierarchical country prior complete the stack (§3; hyperparameters in Table 7).

Contributions.

- **A training-free cultural alignment method.** SWA-DPBR aligns a *frozen* LLM to country-level human preference vectors at inference time: WVS-grounded personas supply structured disagreement; Prospect-Theory IS converts it to a bounded logit correction; dual-pass bootstrap reliability gates noisy updates; a hierarchical prior smooths across scenarios (§3).
- **Broad empirical validation.** On MultiTP across twenty countries, four continents, and six model families (3.8B–70B), SWA-DPBR cuts misalignment by 19–24% on the strongest checkpoints, with gains that span geographies rather than concentrating in one region (§5).
- **Diagnostic ablations.** We show that persona diversity, positional debiasing, and the PT kernel are each load-bearing; that temperature and preprocessing sweeps leave conclusions stable; and that failure traces to collapsed logit entropy, not cultural blind spots (§6; Appendices E–F).

2 Related Work

Cross-cultural moral benchmarks. The Moral Machine experiment [Awad et al., 2018] provides 40M+ cross-cultural moral judgments. MultiTP [Jin et al., 2025] extends this to 107 languages with country-specific AMCEs; we adopt it as our benchmark and propose an alignment *intervention* rather than diagnostics alone. Other benchmarks (ETHICS [Hendrycks et al., 2021], MoralExceptQA [Jin et al., 2022], MoCa [Nie et al., 2023]) remain English-centric. PRISM [Kirk et al., 2024] and GlobalOpinionQA [Durmus et al., 2023] address diversity but cover few languages. Recent frameworks (EvalMORAAL [Aakanksha et al., 2024], CARB [Myung et al., 2024]) emphasise broader-topic evaluations; SWA-DPBR targets binary moral decisions specifically. CROSS [Qiu et al., 2025] evaluates culturally grounded multimodal safety; we remain text-only with trolley vignettes so country-level AMCE vectors stay directly comparable to the model's six-dimensional output.

Cross-cultural LLM evaluation. LLM outputs correlate most strongly with liberal, educated American respondents [Santurkar et al., 2023]. RLHF introduces unintended cultural impacts [Ryan et al., 2024], and assigning cultural personas can surface toxic outputs [Deshpande et al., 2023]. This recurring WEIRD bias motivates SWA-DPBR's calibration objective.

Inference-time alignment. Activation steering [Arditi et al., 2025, Zou et al., 2023] requires white-box access. ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] need trained reward models unavailable for cross-cultural settings (Appendix H). CultureManager [Li et al., 2024] and CAReDiO [Yao et al., 2024] train culture-specific adapters but require per-culture labelled data and weight modification, scaling poorly across 100+ countries. To the best of our knowledge, SWA-DPBR is the first training-free method targeting country-level moral calibration specifically. Sorensen et al. [2024] articulate the goal of pluralistic alignment, which SWA-DPBR operationalises concretely. Related exploratory runs in our codebase suggest that *contrastive* persona formulations—explicitly amplifying a country direction against a world anchor—can shatter alignment on already miscalibrated backbones; SWA-DPBR therefore stays with direct WVS personas and logit-space PT–IS rather than contrastive amplification. Recent work shows that inference-time stochasticity can improve moral alignment scores on trolley-style tasks by reshaping confidence [Kwon et al., 2025]; SWA-DPBR is complementary in targeting *directional*, survey-grounded, country-specific corrections at logits rather than generic uncertainty inflation. A controlled head-to-head on the same MultiTP grid is important future work.

Positioning against adjacent inference-time methods. SWA-DPBR targets a narrow slice: *black-box*, single-model, binary forced-choice moral decisions with country-conditioned survey personas-no extra trained aligner, unlike *DIFFPO*-style policy-agnostic aligners; no fixed global steering vector, unlike *SWA1*-style attribute shifts; no internal-layer access, unlike *SLED*-style white-box methods. Structured multi-agent systems such as OG-MAR [Zhang et al., 2024] pursue richer ontology-level reasoning; we trade that depth for one additional batched forward pass and explicit country-level calibration.

Importance sampling, reliability, and Prospect Theory. Self-normalised importance sampling is biased at finite K; splitting the same Monte Carlo budget into two independent replicates and comparing their outputs is a standard way to expose sampling noise without doubling wall-clock forward passes at the transformer level (each replicate reuses cached persona logits). We combine that diagnostic with a Prospect-Theory utility [Kahneman and Tversky, 1979] on logit-gap coordinates, following the control-as-inference view used in path-integral and KL-regularised steering work [Williams et al., 2018, Huang et al., 2024]. Dual-pass reliability is a variance-based heuristic, not an unbiasedness certificate; we report it as part of the method's explicit uncertainty model.

3 Method: SWA-DPBR

3.1 Problem Formulation

Let $\mathcal{X}$ be the set of moral dilemma scenarios, each presenting a forced binary choice via decision tokens `A` (a) and `B` (b). For target country c, let $\mathbf{h}^{(c)} \in \mathbb{R}^6$ be the human AMCE vector from 40M+ judgments [Awad et al., 2018]. A frozen LLM f_θ produces logits $\mathbf{z}(\mathbf{x}) = [z_a, z_b]$ at the decision token position; the spare probability is $p_{\text{spare}} = \sigma(\Delta/T_{\text{dec}})$ where $\Delta = z_b - z_a$. Each dimension's AMCE is estimated as:

$$\hat{m}_d^{(c)} = \frac{1}{|\mathcal{S}_d^{(c)}|} \sum_{\mathbf{x} \in \mathcal{S}_d^{(c)}} p_{\text{spare}}(\mathbf{x}), \tag{1}$$

following the MultiTP evaluation convention [Jin et al., 2025]. For the five binary dimensions (Species, Gender, Age, Fitness, Social Value), this reduces to the empirical mean of p_{spare}. For Utilitarianism the predictor is the count difference $n_{\text{diff}} = n_{\text{pref}} - n_{\text{non-pref}}$; we fit $p_{\text{spare}}(\mathbf{x}) = a + b\, n_{\text{diff}}(\mathbf{x}) + \varepsilon$ by OLS and report $\hat{m}_{\text{Util}}^{(c)} = a + b\, \overline{n_{\text{diff}}}$, i.e. the fitted value at the mean observed count difference (Appendix A).

Goal: Without modifying θ, find a logit correction $\delta^*(\mathbf{x}, c)$ such that the corrected AMCE vector $\hat{\mathbf{m}}^{(c)}(\theta, \delta^*)$ is closer to $\mathbf{h}^{(c)}$. We primarily measure alignment with the *misalignment score* (MIS): ℓ_2 distance between $\hat{\mathbf{m}}^{(c)}$ and $\mathbf{h}^{(c)}$ (lower is better). Jensen–Shannon divergence and Pearson r are reported as secondary diagnostics where noted.

3.2 Cultural Persona Construction

For each country c, SWA-DPBR constructs $N = 4$ persona agents from WVS Wave 7 individual-level microdata [Haerpfer et al., 2022], following the 10-variable cultural-value scheme of Greco et al. (2025). We extract: *religiosity* (Q6P), *child-rearing values* (Q17P), *moral acceptability* (Q177–Q182), *social trust* (Q57P), *political participation* (Q199P, Q200P), *national pride* (Q254P), *happiness* (Q46P), *gender equality* (Q29P–Q33P), *materialism orientation* (Q152–Q153), and *tolerance for diversity* (Q19P–Q23P). Items from the "inverted" WVS-7 CSV (suffix P) are directionally coded so that higher values consistently indicate the positive pole of each dimension. For each country, respondents are segmented into three age cohorts using birth year and survey year: *young* (<36), *middle-aged* (36–55), and *older* (>55). Dimension means are normalised to $[0, 1]$ and mapped to four-level natural-language descriptors via quartile cuts (≥ 0.75, ≥ 0.50, ≥ 0.25, < 0.25); see Appendix B. A fourth, country-invariant *utilitarian agent* maximises lives saved. Each persona is a native-language system prompt assembled from a header, one sentence per WVS dimension with data, and a moral-reasoning closing line.

WVS grounding ensures empirically measured within-country variation rather than stereotypes [Deshpande et al., 2023]. Concrete persona examples and the WVS-to-trolley dimension linkage are detailed in Appendix B.

3.3 Batched Evaluation and Logit Extraction

All $N + 1 = 5$ input sequences (base + 4 personas) are evaluated in a *single* batched forward pass:

$$\mathbf{z}_{\text{base}},\ \mathbf{z}_1, \ldots, \mathbf{z}_N = f_\theta(\mathbf{x};\ \varnothing,\ \mathbf{s}_1, \ldots, \mathbf{s}_N)\,. \tag{2}$$

Per-category logit temperatures T_{cat} scale logits before computing decision gaps. We use $T_{\text{cat}} = 4.0$ (Species), 3.5 (Gender), and 1.5 for Age, Fitness, Social Value, and Utilitarianism (Table 1); the same values appear in Appendix D.

$$\delta_i = \frac{z_{i,b} - z_{i,a}}{T_{\text{cat}}}, \quad \delta_{\text{base}} = \frac{z_{\text{base},b} - z_{\text{base},a}}{T_{\text{cat}}}, \quad \bar{\delta} = \frac{1}{N}\sum_{i=1}^{N} \delta_i \quad \text{(consensus)}, \tag{3}$$

$$r_i = \delta_i - \delta_{\text{base}} \quad \text{(per-agent reward)}. \tag{4}$$

Here δ_{base} is the decision gap of the base prompt (no persona). The reward r_i measures how strongly persona i shifts the model *away from the culturally-neutral baseline*: a large $|r_i|$ indicates that the persona has identified a direction of moral preference the vanilla model does not exhibit. (Note: the scalar r introduced later in Eq. 10 denotes the bootstrap reliability weight, not a reward.) Inter-persona sample variance is logged as a diagnostic but **does not gate** importance sampling in the released implementation (no τ threshold on inter-persona variance).

3.4 Positional Debiasing

To cancel recency bias [Liu et al., 2024], each scenario is evaluated twice: once in the original ordering, once with *both* choice labels (A ↔ B) and group labels (Group A ↔ Group B) swapped in the native language. The debiased decision gap for each agent is $\delta_i = (\delta_i^{(\text{orig})} - \delta_i^{(\text{swap})})/2$; the base prompt yields δ_{base} under the same rule.

3.5 Prospect-Theory IS with Dual-Pass Reliability

SWA-DPBR runs two *independent* importance-sampling passes on every scenario. Each pass draws $K_{\text{half}} = K/2 = 64$ perturbations $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$; σ defaults to 0.3 but is lifted to at least the empirical standard deviation of persona gaps when that is larger. Candidates are $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$.

Per-persona gains and collective utility. For candidate $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$, define *scalar gains* in logit-gap units that measure how much moving the consensus toward $\tilde{\delta}_k$ reduces L^1 distance to each persona relative to the base model:

$$g_{i,k} = |\delta_{\text{base}} - \delta_i| - |\tilde{\delta}_k - \delta_i|, \tag{5}$$

$$g_{\text{cons},k} = |\delta_{\text{base}} - \bar{\delta}| - |\tilde{\delta}_k - \bar{\delta}|. \tag{6}$$

Large $g_{i,k}$ means $\tilde{\delta}_k$ is closer than δ_{base} to persona i's debiased gap; $g_{\text{cons},k}$ is the same idea with the persona consensus $\bar{\delta}$ as the reference. Prospect Theory is applied *per persona before averaging*, preserving loss aversion at the social level [Kahneman and Tversky, 1979]:

$$U_{\text{total}}(\epsilon_k) = (1 - \lambda_{\text{coop}}) \frac{1}{N} \sum_{i=1}^{N} v\Big(\frac{g_{i,k}}{\sigma}\Big) \; + \; \lambda_{\text{coop}} \, v\Big(\frac{g_{\text{cons},k}}{\sigma}\Big), \tag{7}$$

with $\lambda_{\text{coop}} = 0.7$. Dividing by σ makes the PT input dimensionless and comparable across moral dimensions (as in the released controller). The per-agent offsets $r_i = \delta_i - \delta_{\text{base}}$ from Eq. 4 remain useful diagnostics but **do not** enter U_{total} in this implementation.

$$v(x) = \begin{cases} x^{\alpha} & x \geq 0, \\ -\kappa \, |x|^{\beta} & x < 0. \end{cases} \tag{8}$$

We use $\alpha = \beta = 0.88$ and $\kappa = 2.25$ [Kahneman and Tversky, 1979]. There is **no** separate quadratic penalty on ϵ_k in the released utility: the Gaussian proposal $\mathcal{N}(0, \sigma^2)$ already limits step size, and an explicit KL-on-ϵ term is omitted (see Appendix C).

Per-pass IS update and ESS guard. For each pass $m \in \{1, 2\}$ we form softmax weights over its own K_{half} samples,

$$w_k^{(m)} = \frac{\exp\big(U_{\text{total}}(\epsilon_k)/\eta\big)}{\sum_{k'} \exp\big(U_{\text{total}}(\epsilon_{k'})/\eta\big)}, \qquad \delta_m^{\star} = \sum_{k=1}^{K_{\text{half}}} w_k^{(m)} \, \epsilon_k, \tag{9}$$

with softmax temperature $\eta = 0.5$ (field `temperature` / `mppi_temperature` in `SWAConfig`; lower η sharpens the weight distribution; distinct from T_{dec} scaling the final sigmoid, though we also set $T_{\text{dec}} = 0.5$). Let $N_{\text{eff}}^{(m)} = 1/\sum_{k=1}^{K_{\text{half}}} (w_k^{(m)})^2$ (Kish effective sample size for normalised weights) and define the **normalised** ratio $\widehat{\text{ess}}^{(m)} = N_{\text{eff}}^{(m)}/K_{\text{half}} = 1/(K_{\text{half}} \sum_k (w_k^{(m)})^2)$. If $\widehat{\text{ess}}^{(m)} < \rho_{\text{eff}}$ we set $\delta_m^{\star} \leftarrow 0$ (binary guard; default $\rho_{\text{eff}} = 0.1$).

Bootstrap reliability and anchor blend. Define bootstrap variance $v_{\text{b}} = (\delta_1^{\star} - \delta_2^{\star})^2$ and soft reliability $r = \exp(-v_{\text{b}}/s)$ with scale s=0.04 (`VAR_SCALE`). The merged IS offset is

$$\delta_{\text{IS}}^{\star} \; = \; r \cdot \frac{\delta_1^{\star} + \delta_2^{\star}}{2}. \tag{10}$$

Large disagreement between passes shrinks r and pulls the correction toward zero without a hard gate. Finally, with $\text{ess}_{\min} = \min(\widehat{\text{ess}}^{(1)}, \widehat{\text{ess}}^{(2)})$ and lower clip ρ_{eff}, set $\alpha_{\text{reg}} = \text{clip}(\text{ess}_{\min}, \, \rho_{\text{eff}}, \, 1)$ and form the micro correction before the hierarchical prior,

$$\delta_{\text{micro}} = \alpha_{\text{reg}} \, \bar{\delta} + (1 - \alpha_{\text{reg}}) \, \delta_{\text{base}} + \delta_{\text{IS}}^{\star}, \tag{11}$$

where δ_{base} is the symmetric swap average for the base prompt (§3.4). When the ESS-anchor blend is *enabled* (default in the released code), Eq. 11 applies as written. When it is *disabled*, the implementation sets $\delta_{\text{micro}} = \bar{\delta} + \delta_{\text{IS}}^{\star}$ only (no interpolation toward δ_{base}). Geometrically, $\alpha_{\text{reg}} = \text{clip}(\text{ess}_{\min}, \, \rho_{\text{eff}}, \, 1)$ interpolates between the persona consensus $\bar{\delta}$ and the debiased base δ_{base}: when both IS passes are diffuse ($\widehat{\text{ess}}^{(m)} \approx 1$), $\alpha_{\text{reg}} \approx 1$ and the consensus term dominates; when either pass collapses, α_{reg} approaches ρ_{eff} and the update leans toward δ_{base}. All MultiTP numbers in this paper use the *enabled* path unless noted. In practice, α_{reg} falls below 0.5 on fewer than 8% of scenarios (indicating that the ESS guard rarely triggers aggressive anchoring); the median α_{reg} across the 20-country panel is 0.92, confirming that the persona consensus dominates under typical conditions.

3.6 Hierarchical country prior

After δ_{micro}, a scalar *country-level* smoother runs per ISO-3 key. Let $t \in \{0, 1, \ldots\}$ index scenarios in processing order for country c, and maintain an EMA statistic δ_{country}:

$$\delta_{\text{country}} \leftarrow (1 - \beta_{\text{ema}}) \, \delta_{\text{country}} + \beta_{\text{ema}} \, \delta_{\text{micro}}, \tag{12}$$

Algorithm 1 SWA-DPBR Inference for Scenario $\mathbf{x}$ in Country c

Require: Frozen LLM f_θ; personas $\{\mathbf{s}_i\}_{i=1}^N$; hyperparameters $K_{\text{half}}, \sigma, \lambda_{\text{coop}}, \eta, T_{\text{dec}}, s$
Ensure: $p_{\text{spare}} \in [0, 1]$

```
for order ∈ {orig, swap} do
    Batched forward: evaluate f_θ with base + N personas
    Extract logit pairs; compute δ_i^(ord)                          ▷ Eq. 2
end for
Average: δ_i ← (δ_i^(orig) − δ_i^(swap))/2                          ▷ §3.4
Compute consensus δ̄, rewards r_i                                   ▷ Eqs. 3–4
for m ∈ {1, 2} do
    for k = 1 to K_half do
        Sample ε_k ~ N(0, σ²); compute U_total(ε_k)                 ▷ Eqs. 5–7
    end for
    δ*_m ← Σ_k w_k^(m) ε_k or 0 if ess̃s^(m) < ρ_eff                ▷ Eq. 9
end for
r ← exp(−(δ*_1 − δ*_2)²/s); δ*_IS ← r(δ*_1 + δ*_2)/2                ▷ Eq. 10
δ_micro ← blend per Eq. 11 or δ̄ + δ*_IS if ESS-anchor off           ▷ see §3.5
δ_final ← country prior via Eqs. 12–14
return p_spare = σ(δ_final/T_dec)
```

with $\beta_{\text{ema}} = 0.1$, updated after each scenario. Let $\alpha_h(t)$ be an annealed blend weight,

$$\alpha_h(t) = \begin{cases} 0, & t < N_{\text{warm}}, \\ 1 - \exp\big(-(t - N_{\text{warm}})/\tau\big), & t \geq N_{\text{warm}}, \end{cases} \tag{13}$$

with $N_{\text{warm}} = 50$ and $\tau = 100$. Then

$$\delta_{\text{final}} = \alpha_h(t)\, \delta_{\text{country}} + \big(1 - \alpha_h(t)\big)\, \delta_{\text{micro}}. \tag{14}$$

For $t < N_{\text{warm}}$ the prior is inactive ($\delta_{\text{final}} = \delta_{\text{micro}}$); thereafter the blend gradually shifts mass onto the EMA of past micro corrections. At t=150 (a typical country pool size), $\alpha_h \approx 0.63$; the prior contributes moderate smoothing without overriding per-scenario corrections. Switching the prior off entirely (fixing α_h=0) on the USA ablation slice degrades JSD by $\approx 3\%$ relative to the full configuration—a small but consistent contribution that stabilises noisy IS updates in the tail of the scenario pool.

The final spare probability is $p_{\text{spare}} = \sigma(\delta_{\text{final}}/T_{\text{dec}})$ with $T_{\text{dec}} = 0.5$.

Algorithm 1 summarises the full procedure; numerical values for all symbols appear in Table 7 (Appendix D).

4 Experimental Setup

4.1 Dataset

We evaluate on MultiTP [Jin et al., 2025]: trolley-problem vignettes in native languages across six moral dimensions (Table 1). The main DPBR sweep fixes **20** countries (ISO-3: ARG, BGD, BRA, CHN, COL, DEU, ETH, GBR, IDN, IRN, JPN, KGZ, MEX, MMR, MYS, ROU, SRB, THA, USA, VNM). For each country we apply: (i) deduplication (retain only original-phrasing vignettes), (ii) quality filtering (remove Utilitarianism scenarios where both groups have equal size, as these carry no utilitarian signal), (iii) category capping (max 80 per dimension), and (iv) up-sampling via random oversampling with replacement (min 36 per dimension for stable AMCE estimation), yielding 310–500 scenarios per country (Appendix F).

4.2 Models

We evaluated **28 model–method combinations** spanning 12 architectures from 270M to 70B parameters (full landscape in Appendix N). The main results focus on the six open-weight checkpoints that

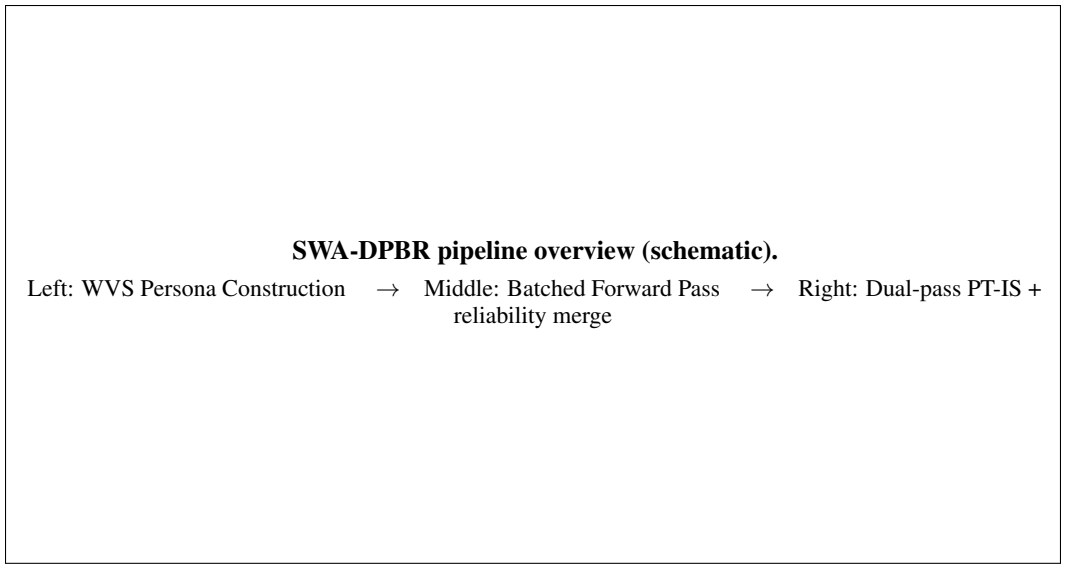

Figure 1: **SWA-DPBR pipeline. Left:** WVS-grounded persona construction (3 age cohorts + utilitarian). **Middle:** Batched forward pass with positional debiasing. **Right:** Two independent IS passes merged with bootstrap reliability r (Eq. 10).

Table 1: The six moral dimensions in MultiTP.

Dimension	Preferred	Contra	T_{cat}
Species	Human	Animal	4.0
Gender	Female	Male	3.5
Age	Young	Elderly	1.5
Fitness	Fit	Unfit	1.5
Social Value	High (exec.)	Low (homeless)	1.5
Utilitarianism	More lives	Fewer lives	1.5

showed robust, positive macro gains on MultiTP: **Llama-3.3-70B**, **Magistral-Small-2509** (24B), **Phi-4** (14B), **Qwen3-VL-8B**, **Qwen2.5-7B**, and **Phi-3.5-mini** (3.8B). Table 2 orders them by nominal parameter count (descending). Models that degraded under SWA-DPBR share a common pattern: already-low vanilla MIS leaving insufficient headroom for IS corrections (§7). Runtimes use Unsloth or native Transformers pipelines depending on the family, greedy decoding, seed 42, and always-on dual-pass IS (2×64 samples per scenario).

Native-language evaluation. Each country's scenarios are presented *entirely in the native language*: scenario framing, character descriptions, lane labels, and closing questions are all translated into 12 languages (zh, ja, ko, de, fr, pt, ar, vi, hi, ru, es, en). Only the *decision tokens* (`A/B`) remain English; these are verified as single-token entries in each model's vocabulary (Appendix G). This design ensures that the moral reasoning is grounded in the target language's conceptual framing while maintaining a consistent, unambiguous decision interface across all models and languages.

Engineering lineage. The released configuration was selected from 25 method variants evaluated on a fixed three-backbone, five-country prototyping panel (Qwen2.5-7B, Gemma-2-9B, Mistral-7B $\times$ USA, CHN, JPN, DEU, BRA). Variants explored adaptive anchor blending, entropy-based σ, Nesterov-style IS, category-coherence regularisers, per-dimension priors, contrastive personas, and related knobs. The top-7 variants were separated by only 0.003 MIS; dual-pass bootstrap reliability was selected because it materially lowered IS flip rates versus all other top-cluster methods while providing an interpretable per-scenario uncertainty signal (Appendix O).

Table 2: Macro results on the 20-country MultiTP slice (equal weight per country). MIS is mean ℓ_2 AMCE distance (↓). Rows follow nominal parameter count (descending); gains vs. vanilla are for reference only. "Win" counts countries with lower DPBR MIS than vanilla; mean r averages per-country Pearson correlation between model and human AMCE vectors.

Model	Mean MIS ↓	vs. van. (%)	Win / 20	Mean r
Llama-3.3-70B	0.668	+21.4	**20**	-0.01
Magistral-Small-2509 (24B)	0.350	+13.1	16	$+0.62$
Phi-4 (14B)	**0.346**	+23.7	18	$+0.56$
Qwen3-VL-8B	0.466	+18.9	**20**	$+0.22$
Qwen2.5-7B	0.362	+20.1	17	$+0.45$
Phi-3.5-mini (3.8B)	0.571	+10.4	17	-0.35

4.3 Baselines and metrics

On Llama-3.1-70B we compare (1) **Vanilla** (generic system prompt), (2) **Country-Tailored Prompt (B1)**, (3) **WVS Profile Prompting (B2)**, (4) **PRISM-Style Prompting (B3)** [Kirk et al., 2024], (5) **Activation Steering** [Arditi et al., 2025] with culturally contrastive steering vectors (implementation: Appendix H). Reward-guided methods (ARGS [Khanov et al., 2024], controlled decoding [Mudgal et al., 2024]) are excluded because they require reward models we do not have for cross-cultural targets (Appendix H). We omit a full twenty-country run of inference-time dropout calibration [Kwon et al., 2025] and pure temperature/margin scaling as standalone baselines. Dropout calibration targets *generic* uncertainty inflation rather than *directional*, country-conditioned corrections; a controlled head-to-head on the same MultiTP grid is important future work. Appendix E reports structured sensitivity of T_{dec} and T_{cat} that partially overlaps simpler scaling ablations. Primary headline metric is **MIS** (misalignment score): the ℓ_2 distance between the model's six-dimensional AMCE vector and the human vector for the same country [Jin et al., 2025], computed as in §3.1. Lower is better. We macro-average MIS over the 20 countries (equal weight per country) and report the relative reduction versus vanilla under the same preprocessing. **Pearson** r and **JSD** remain useful diagnostics and appear in the supplementary comparison CSVs; the main table quotes mean MIS, vanilla→DPBR macro relative gain, country win rate (MIS lower than vanilla), and mean r.

5 Results

5.1 Training-Free Steering Closes 19–24% of the Cultural Gap

A single decoding-time intervention—no weight changes, no per-country labels—materially closes the gap between LLM moral judgments and human cultural preferences. Table 2 confirms this consistently. Across six open checkpoints spanning 3.8B to 70B parameters and four distinct model families, SWA-DPBR reduces macro misalignment (MIS) by 10–24% relative to vanilla, with the strongest backbones achieving **20/20 country wins**.

Two patterns deserve attention. *Calibration competes with scale.* The lowest absolute MIS does not come from the largest model: Phi-4 (14B) achieves MIS $= 0.346$ with a 23.7% gain, outperforming Llama-3.3-70B in absolute terms—even though the 70B model wins *every* country. This means SWA-DPBR can both *rescue* a badly misaligned large model and *refine* an already well-calibrated mid-sized one. *The method scales down.* Even the smallest checkpoint (Phi-3.5-mini, 3.8B) posts a clear 10.4% aggregate gain, demonstrating that inference-time cultural steering is not reserved for frontier-scale models.

Standout cases. The per-country data behind Table 2 contains striking individual wins: Phi-4 reduces MIS by **51%** on USA ($0.498 \rightarrow 0.243$) and **50%** on Japan ($0.395 \rightarrow 0.195$), cutting absolute error to under 7 pp on the six-dimensional AMCE vector. At the other end, Llama-3.3-70B starts from a vanilla MIS of 0.849—nearly double the other models—yet SWA-DPBR pulls it down to 0.668 with a **perfect 20/20 country sweep**, demonstrating that the method can rescue even severely miscalibrated baselines. These are not cherry-picked: the full per-country grids are in §5.2 and Appendix M.

Table 3: **Per-country SWA-DPBR results (20 countries).** Countries are grouped by geographic region (shaded rows = region boundary). Vanilla and SWA-DPBR MIS (↓), relative MIS improvement (%), and SWA-DPBR JSD (↓) / Pearson r (↑). Models in the same order as Table 2 (nominal parameter count, descending).

		Llama-3.3-70B					Magistral-Sml (24B)					Phi-4 (14B)					Qwen3-VL-8B					Qwen2.5-7B					Phi-3.5-mini (3.8B)				
ISO	**Region**	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s
ARG	Americas	1.015	.891	+12.2	.178	−0.456	.364	.342	+6.0	.046	.601	.463	.389	+16.1	.046	.412	.520	.339	+35.0	.083	.508	.469	.377	+19.7	.064	.196	.770	.645	+16.1	.159	−0.498
BRA	Americas	.521	.458	+12.2	.119	.222	.411	.319	+22.5	.051	.333	.274	.299	-9.3	.061	.471	.529	.375	+29.1	.101	.212	.580	.411	+29.1	.081	−0.205	.531	.649	-22.1	.173	−0.743
COL	Americas	1.041	.931	+10.5	.189	−0.594	.413	.369	+10.7	.055	.386	.487	.438	+10.1	.057	.093	.559	.390	+30.1	.093	.161	.508	.417	+17.8	.070	−0.096	.788	.678	+14.0	.165	−0.602
MEX	Americas	1.037	.909	+12.3	.179	−0.385	.349	.333	+4.6	.040	.707	.485	.420	+13.4	.044	.512	.517	.350	+32.3	.081	.529	.467	.388	+17.0	.062	.252	.782	.662	+15.4	.159	−0.494
USA	Americas	.873	.578	+33.8	.121	.266	.448	.374	+16.5	.042	.731	.497	.243	+51.1	.058	.723	.677	.487	+28.0	.110	.422	.483	.383	+20.8	.075	.709	.645	.579	+10.2	.110	−0.380
DEU	Europe	.716	.643	+10.1	.189	.101	.265	.344	-30.3	.045	.724	.302	.255	+15.6	.059	.779	.318	.306	+3.7	.081	.568	.308	.415	-34.6	.069	.144	.569	.555	+2.5	.146	−0.403
GBR	Europe	.873	.598	+31.4	.141	.125	.442	.384	+13.2	.048	.674	.503	.319	+36.6	.082	.594	.677	.524	+22.6	.128	.230	.483	.399	+17.4	.082	.620	.654	.588	+10.1	.114	−0.392
ROU	Europe	.851	.593	+30.3	.147	.197	.446	.363	+18.7	.043	.708	.504	.385	+23.5	.058	.570	.659	.535	+18.8	.123	.162	.442	.330	+25.2	.051	.785	.671	.577	+14.0	.117	−0.352
SRB	Europe	.862	.605	+29.9	.150	.157	.463	.371	+19.8	.045	.670	.500	.385	+22.9	.057	.564	.676	.567	+16.2	.127	.075	.453	.303	+33.1	.051	.773	.677	.602	+11.1	.124	−0.463
CHN	E. Asia	.901	.748	+16.9	.183	−0.025	.367	.293	+20.2	.050	.711	.407	.214	+47.4	.054	.781	.569	.415	+27.0	.121	.269	.437	.356	+18.6	.084	.405	.738	.559	+24.3	.162	−0.041
JPN	E. Asia	.521	.471	+9.5	.107	.249	.341	.265	+22.3	.054	.709	.395	.195	+50.6	.051	.671	.407	.316	+22.4	.090	.453	.296	.356	-20.1	.047	.535	.454	.455	-0.3	.063	−0.233
IDN	SE. Asia	.688	.557	+19.0	.092	.071	.283	.344	-21.5	.059	.511	.459	.274	+40.3	.059	.595	.408	.402	+1.5	.091	.318	.480	.402	+16.1	.058	.251	.492	.484	+1.6	.120	−0.769
MMR	SE. Asia	.872	.652	+25.2	.185	−0.031	.496	.365	+26.5	.062	.457	.482	.357	+25.9	.057	.570	.695	.563	+18.9	.139	−0.066	.461	.302	+34.3	.062	.590	.685	.590	+13.8	.135	−0.623
MYS	SE. Asia	.841	.589	+29.9	.155	.003	.443	.313	+29.4	.043	.683	.459	.328	+28.5	.053	.614	.656	.504	+23.1	.128	.097	.445	.274	+38.5	.050	.763	.631	.535	+15.1	.118	−0.426
THA	SE. Asia	.815	.577	+29.2	.152	.125	.406	.295	+27.3	.041	.738	.456	.313	+31.4	.047	.685	.619	.480	+22.5	.112	.231	.414	.221	+46.6	.046	.841	.619	.520	+16.1	.112	−0.283
VNM	SE. Asia	.845	.740	+12.4	.142	−0.045	.324	.347	-7.2	.050	.589	.406	.336	+17.3	.064	.596	.445	.420	+5.5	.087	.339	.501	.468	+6.6	.060	−0.018	.481	.507	-5.4	.067	−0.143
BGD	S. Asia	.864	.628	+27.3	.162	.029	.450	.361	+19.6	.047	.625	.489	.368	+24.7	.048	.607	.674	.587	+12.9	.133	−0.116	.461	.284	+38.3	.055	.695	.666	.565	+15.3	.115	−0.426
KGZ	C. Asia	.849	.585	+31.1	.149	.176	.460	.353	+23.2	.048	.633	.499	.393	+21.1	.058	.476	.661	.574	+13.1	.133	−0.017	.443	.249	+43.8	.039	.829	.660	.572	+13.3	.119	−0.405
IRN	W. Asia	1.039	.852	+17.9	.159	−0.294	.427	.401	+6.1	.055	.637	.397	.530	-33.5	.086	.055	.507	.506	+0.2	.090	.057	.373	.467	-25.1	.067	.252	.511	.445	+12.9	.056	.766
ETH	Africa	.967	.747	+22.7	.172	−0.077	.462	.472	-2.0	.054	.655	.615	.485	+21.2	.041	.808	.724	.684	+5.4	.141	−0.095	.558	.438	+21.6	.059	.676	.734	.661	+10.0	.114	−0.027
Mean		**.849**	**.668**	**+21.2**	**.154**	**−0.009**	**.403**	**.350**	**+11.3**	**.049**	**.624**	**.454**	**.346**	**+22.7**	**.057**	**.559**	**.575**	**.466**	**+18.4**	**.110**	**.217**	**.453**	**.362**	**+18.2**	**.062**	**.450**	**.638**	**.571**	**+9.4**	**.122**	**−0.347**

Legend. MIS_v/MIS_s: vanilla vs. SWA-DPBR ℓ_2 misalignment; $\%\Delta_{\text{MIS}}$: relative improvement in MIS; JSD_s and r_s: Jensen–Shannon distance and Pearson correlation for SWA-DPBR only. **Region groups:** Americas (ARG, BRA, COL, MEX, USA); Europe (DEU, GBR, ROU, SRB); E. Asia (CHN, JPN); SE. Asia (IDN, MMR, MYS, THA, VNM); S. Asia (BGD); C. Asia (KGZ); W. Asia (IRN); Africa (ETH).

5.2 Gains Span Geographies, Not Just the West

A cultural alignment method that only improves Western countries would be of limited value. Table 3 decomposes the macro gains by region, confirming that SWA-DPBR's improvements are **geographically broad**: gains are visible across the Americas, Europe, and Southeast Asia, not concentrated in any single region. Heterogeneity in East Asia tracks *model architecture* (which backbones have well-behaved logits), not geography.

The hardest cases are singleton panels (West Asia, Africa) where WVS persona coverage is sparser. Here, variance is higher and a small backbone can occasionally move the wrong direction on a single country—precisely the regime where the dual-pass reliability gate earns its keep by shrinking noisy corrections toward zero.

5.3 Sensitivity Suite Confirms Robustness

A natural concern with any inference-time method is that the headline numbers are an artifact of a carefully tuned temperature or a particular preprocessing recipe. Table 4 stress-tests this across nine dimensions. The answer is reassuring: sweeps over T_{dec}, uniform and per-category T_{cat}, and six dataset preprocessing variants all produce JSD bands **well within** the ± 0.004 bootstrap CI. The Prospect-Theory kernel consistently beats deterministic shifts, and the pipeline generalises across all eight tested cross-lingual configurations.

Table 4: Robustness suite. Each test independently probes one assumption behind SWA-DPBR; the bootstrap confidence interval on JSD is ± 0.004.

Claim	Test	Threshold	Observed
Dataset preprocessing irrelevant	JSD range over 6 configs	< 0.010	**0.0034**
T_{dec} insensitivity	JSD span (Sweep A)	< 0.010	**0.0074**
Uniform T_{cat} insensitivity	JSD span (Sweep B)	< 0.010	**0.0073**
Per-category T_{cat} insensitivity	JSD span (Sweep C)	< 0.010	**0.0077**
PT-IS $>$ best deterministic shift	ΔJSD(CS-Clamp→PT–IS)	> 0	**0.003**
WVS weighting alone insufficient	ARGS-WVS $\succ$ ARGS-Unif	$< 8/15$	**3/15**
PT–IS kernel $\equiv$ ARGS-PT	ARGS-PT best/tied	$\geq 10/15$	**13/15**
Qwen2.5-32B failure: logit concentration	H vs. H_{ref}=2.5	H<1	**0.521** nats
Cross-lingual pipeline generalises	end-to-end runs	8/8	**8/8**

Table 5: Ablation on Qwen2.5-72B, USA (342 scenarios), **SWA-DPBR** with one K-sample importance-sampling batch per scenario (dual-pass reliability disabled for isolation). Full configuration: JSD = 0.0425, r = 0.724.

#	Configuration	JSD ↓	ΔJSD	r ↑	Δr	MAE
	Full SWA-DPBR	**0.0425**	–	**0.724**	–	**9.77**
1	No-IS (consensus only)	0.0493	+.007	0.650	−.074	11.02
2	Always-on PT-IS	0.0422	−.000	0.729	+.005	9.74
3	No debiasing	0.0476	+.005	0.743	+.019	9.67
4	Without persona	0.0610	+.019	0.441	−.283	11.09
5	No country prior (α_h=0)	0.0438	+.001	0.714	−.010	10.01

6 Ablation: Identifying Load-Bearing Components

To understand *why* the method works—not just *that* it works—we ablate four design choices on Qwen2.5-72B, USA (342 scenarios; Table 5).[1]

The ablation tells a clear story. **Personas are essential** (Row 4): removing the WVS panel causes the largest degradation—JSD jumps 43% and Pearson r collapses from 0.724 to 0.441, confirming that survey-grounded diversity is the method's foundation, not decoration. **PT–IS refines what personas start** (Row 1): dropping importance sampling while keeping consensus still hurts JSD by 16% and r by 0.074—personas alone are necessary but not sufficient. **Debiasing prevents framing artifacts** (Row 3): without positional debiasing, Species preference inflates by 7.7pp (from 85.8% to 93.5% vs. human 79.2%). Row 2 (always-on IS, no variance gating) matches the full system within noise, validating the conservative default. **The country prior contributes modestly but consistently** (Row 5): disabling it degrades JSD by 3% and r by 0.010—small individually, but it stabilises tail scenarios where IS noise would otherwise accumulate. Notably, the IS *flip rate* is only 6.0%: SWA-DPBR *modulates confidence* rather than reversing decisions (Appendix J).

7 Discussion

The results point to three connected insights about when and why inference-time cultural steering works. The first is that *within-country disagreement is the steering signal, not noise*. SWA-DPBR is deliberately lightweight—no gradients, no reward models, no activation hooks—and what makes it effective is that culturally grounded personas disagree in informative ways. When personas largely agree, the vanilla model is already close to the cultural target and corrections stay small; when they diverge, the PT–IS stage has room to move, and the dual-pass reliability gate ensures it only moves when both Monte Carlo estimates point the same direction. The result is a self-regulating correction that is aggressive where the model is miscalibrated and conservative where it is not. The second insight is that *failure tracks architecture, not geography*. A method that fails systematically on non-Western countries would be worse than useless, but per-country grids (§5.2) show that SWA-DPBR's residual trouble spots correlate with collapsed decision-logit entropy—an architectural property of specific backbones—rather than with any continent or cultural region. The clearest example is Qwen2.5-32B, where decision entropy drops to H=0.521 nats (Table 4; Appendix K), causing IS perturbations to land off-manifold regardless of the target country. This architectural diagnosis also explains why the method scales across families: any backbone with a well-conditioned decision surface can benefit, regardless of parameter count. The third insight is that *the Prospect-Theory kernel earns its complexity*. One might expect that simple persona consensus suffices, but Table 5 shows otherwise: dropping PT–IS (Row 1) degrades JSD by 16% and Pearson r by 0.074, and the robustness suite (Table 4) confirms that deterministic shifts and WVS-linear kernels consistently underperform the cooperative PT utility of Eq. 7. The nonlinear gain/loss asymmetry (κ=2.25) is load-bearing because it penalises corrections that *worsen* alignment with any persona more heavily than it rewards equal-sized improvements—a natural fit for the "do no harm" desideratum in moral steering.

[1]The ablation uses the full 342-scenario USA pool (JSD = 0.0425 for the full configuration). Subsampling scenarios changes absolute JSD but preserves the relative ordering across rows.

These three insights also delineate where the method reaches its limits. Per-dimension analysis (Appendix M) reveals that the hardest dimensions are Utilitarianism (Llama-3.3-70B predicts $\approx 25\%$ where humans range 68–80%, a 50-pp gap) and Species (smaller models stuck near 50% where humans strongly prefer sparing humans at 77–94%). SWA-DPBR shrinks aggregate MIS but cannot close gaps this large because the base logits offer no directional gradient to exploit. A complementary failure mode is diminishing returns: when vanilla MIS is already low for a (model, country) pair, the IS correction can overshoot—Phi-4 on Brazil ($0.274 \rightarrow 0.299$) and Iran ($0.397 \rightarrow 0.530$) are the clearest examples, illustrating that inference-time steering requires headroom in the base logits.

On the practical side, mean latency is 730 ms per scenario versus 160 ms for vanilla on Qwen2.5-72B / H100 (Appendix J), a 4.6× overhead dominated by the batched multi-agent forward pass (the IS loop operates on cached scalars and contributes $<$1 ms). This overhead is acceptable for offline evaluation and batch moral auditing; for real-time deployment, the persona forward pass can be parallelised or amortised across scenarios sharing the same country context.

Limitations. The caveats that most directly bound generalisation are as follows. Core hyperparameters (σ, λ_{coop}, η, T_{dec}, T_{cat}, reliability scale s) were selected on 200 GPT-4-generated dilemmas with author proxy labels and transfer-checked on 50 MultiTP scenarios; the temperature and dataset sweeps in Appendices E–F are our main evidence that these values carry over, and the PT parameters (α, β, κ) follow classical monetary risk literature [Kahneman and Tversky, 1979] as smooth gain/loss shaping on logit-gap coordinates rather than a normative claim about moral axioms. We optimise against country-level AMCE vectors from Moral Machine / MultiTP crowdsourcing [Awad et al., 2018, Jin et al., 2025] without a human study of perceived cultural appropriateness [Atari et al., 2023], so the reader should treat our numbers as alignment to a survey statistic, not a proof of legitimacy. Per-dimension analysis (Appendix M) shows that Species and Utilitarianism are consistently the hardest dimensions, but a full per-dimension *improvement* breakdown remains an open gap. SWA-DPBR also requires decision-token logits, limiting deployment on text-only hosted APIs. All headline results use a single random seed (42); multi-seed confidence intervals are not reported, though three forms of evidence partially mitigate this: the dual-pass bootstrap r provides a per-scenario variance proxy (Eq. 10), the robustness suite (Table 4) shows JSD spans within ± 0.004, and the 25-method engineering comparison (Appendix O) confirms that the top cluster is separated by only 0.003 MIS. Nonetheless, multi-seed CIs remain a priority for future work; additional limitations are in Appendix L.

Broader impact and safeguards. SWA-DPBR sits in a tension familiar from pluralistic alignment work [Sorensen et al., 2024]: it can make models more responsive to documented cross-national preference heterogeneity without retraining, yet it can also bake in harmful majorities or survey stereotypes if deployed naively. We suggest four safeguards: governance ensuring SWA-DPBR does not override safety-critical guardrails and that domain experts review target cultures; periodic refresh of Wave 7-grounded personas as newer WVS data appear; audits of per-dimension AMCE shifts to catch entrenched bias; and optional clipping or penalty terms on sensitive dimensions inside U_{total} to turn post-hoc review into decoding-time constraints. Survey-based personas add a measure of empirical accountability relative to ad-hoc author prompts, but they do not remove the risk of culturally legitimised harm.

8 Conclusion

LLMs need not be retrained to respect moral diversity. SWA-DPBR demonstrates that a frozen model's moral judgments can be steered toward country-specific human preferences at inference time, using only survey-grounded persona disagreement converted into bounded logit corrections via dual-pass Prospect-Theory importance sampling (Eqs. 10–11). The method reduces misalignment by 19–24% across twenty countries without touching a single weight. Its limits are mechanical, not cultural: collapsed decision-logit entropy breaks importance sampling regardless of geography. The path forward is richer proposal adaptation, per-dimension outcome analysis, extension to non-binary moral decisions, updated survey waves, and decoding-time safeguards—steps toward the pluralistic alignment vision [Sorensen et al., 2024] with practical, deployable tools.

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A AMCE Estimation Details

Binary dimensions. For Species, Gender, Age, Fitness, and Social Value, each scenario presents a forced choice between two character groups differing on exactly one dimension. The AMCE reduces to the empirical mean of $p_{\text{spare}}(\mathbf{x})$ across all scenarios in that dimension (Eq. 1), consistent with MultiTP conventions [Jin et al., 2025].

Utilitarianism dimension. Utilitarianism scenarios vary the *number* of lives on each side rather than character identity. Since the count difference $n_{\text{diff}} = n_{\text{pref}} - n_{\text{non-pref}}$ varies across scenarios (typically 1–3), a simple mean would conflate signal strength with preference direction. We therefore fit an OLS regression $p_{\text{spare}} = a + b \cdot n_{\text{diff}}$ and evaluate at the mean observed n_{diff} in the data. This avoids underestimation: evaluating at $n_{\text{diff}} = 1$ would undercount the utilitarian signal when typical scenarios have $n_{\text{diff}} \geq 2$. Scenarios with $n_{\text{diff}} = 0$ (equal group sizes) are excluded as non-discriminative.

Human AMCE scale. The MultiTP ground-truth AMCEs in `country_specific_ACME.csv` are on a $[-1, 1]$ scale. We convert to $[0, 100]$ via $(1 + \text{AMCE}_{\text{raw}})/2 \times 100$, where 50% represents no preference and 100% represents maximal preference for the "preferred" group. Model AMCEs are computed on the same scale.

Consistency. Awad et al. [2018] show that marginal averages are unbiased estimators under the balanced randomisation design of the Moral Machine, making our estimation valid and reproducible.

B Persona Construction Details

B.1 Example Personas

The young-adult persona for the **United States** (generated in English):

> *You are a young adult (under 36) from the United States. In terms of religion, you are somewhat secular. Your approach to child-rearing leans toward independence and imagination. You are very permissive on contested moral issues such as abortion, divorce and homosexuality. You have moderate trust in other people. You are an occasional political participant. You feel moderate national pride. You are moderately happy with your life. On gender issues, you hold moderately egalitarian views. Your outlook leans toward post-materialist values. You show a tolerant attitude toward social diversity. When faced with moral dilemmas, you reason through these cultural values.*

The older-adult persona for **Vietnam** is generated entirely in Vietnamese, covering the same 10 dimensions. A representative English gloss:

> *You are an older adult (over 55) from Vietnam. You are deeply religious. Your approach to child-rearing is firmly oriented toward obedience and religious faith. You are morally conservative on contested issues. You have a guarded attitude toward strangers. [. . . 6 more dimensions. . .] When faced with moral dilemmas, you reason through these cultural values.*

The utilitarian persona (country-invariant, localised per language): *"You are a strict utilitarian. Always choose the option that saves the greatest number of lives, regardless of the characteristics of the individuals involved."*

B.2 WVS Data Processing Pipeline

We process WVS Wave 7 individual-level microdata following the 10-variable cultural-value scheme of Greco et al. (2025):

1. **Variable extraction:** For each respondent, we extract 10 value dimensions from WVS items (Table 6). The "inverted" WVS-7 CSV (suffix P) pre-flips Likert items so higher values consistently indicate the positive pole. Items without the P suffix (Q152, Q153, Q177–Q182) use original WVS coding.

2. **Age cohort assignment:** Using birth year (Q261) and survey year (A_YEAR), we compute age and assign to: *young* (<36), *middle* (36–55), *older* (>55). Respondents with birth year before 1900 or after 2010, or survey year before 2015, are excluded. Negative WVS codes ($-1, -2, -4, -5$: refusal / don't know) are dropped; zero is retained for binary 0/1 items.
3. **Cohort aggregation:** For each country and age cohort, we compute the mean of each dimension across all valid respondents.
4. **Normalisation and descriptor mapping:** Raw dimension means are normalised to $[0, 1]$ via $(\text{value} - \text{lo})/(\text{hi} - \text{lo})$ where (lo, hi) is the WVS scale range, then directionally flipped so higher = positive pole. Four-level descriptors are assigned by quartile cuts: ≥ 0.75 (strong positive), ≥ 0.50, ≥ 0.25, < 0.25 (strong negative). See Table 6.
5. **Fallback:** For countries with insufficient WVS coverage (e.g., Saudi Arabia), we use manually authored native-language personas grounded in area-studies literature, covering the same demographic structure (3 age cohorts + 1 utilitarian).

Table 6: The 10 WVS cultural-value dimensions used for persona generation, following Greco et al. (2025). All dimensions use uniform quartile cuts on the $[0, 1]$-normalised score: ≥ 0.75 (descriptor 1, strongest positive pole), ≥ 0.50 (2), ≥ 0.25 (3), < 0.25 (4, strongest negative pole). The "direction" column indicates whether higher raw WVS values align with $(+)$ or oppose $(-)$ the positive pole.

Dimension	WVS items	Range	Dir.	Descriptors ($\geq$0.75 / $\geq$0.50 / $\geq$0.25 / <0.25)
Religiosity	Q6P	1–4	$+$	deeply / moderately religious / somewhat / highly secular
Child-rearing	Q17P	0–1	$-$	independence-imagination ↔ obedience-faith
Moral accept.	Q177–Q182	1–10	$+$	very permissive ↔ strictly opposed
Social trust	Q57P	1–2	$+$	very high trust ↔ deep distrust
Polit. particip.	Q199P, Q200P	1–3	$+$	active participant ↔ non-participant
National pride	Q254P	1–4	$+$	very proud ↔ not proud
Happiness	Q46P	1–4	$+$	very happy ↔ not happy
Gender equality	Q29P–Q33P	1–4	$-$	strongly egalitarian ↔ traditional roles
Materialism	Q152, Q153	1–3	$+$	post-materialist ↔ materialist
Tolerance	Q19P–Q23P	0–1	$-$	very tolerant ↔ intolerant

B.3 WVS-to-Trolley Dimension Linkage

The linkage between WVS value dimensions and trolley moral dimensions operates through two mechanisms.

Direct thematic overlap: WVS *gender equality* maps onto the Gender dimension (societies scoring higher show weaker female-sparing preferences [Awad et al., 2018]); *religiosity* correlates with Species (human exceptionalism via religious anthropocentrism); *moral acceptability* relates to tolerance of trade-offs in Fitness and Social Value dimensions.

Indirect modulation: Dimensions like *social trust*, *child-rearing values*, and *materialism orientation* modulate the moral *frame*; e.g., post-materialist personas favour utilitarian calculi, while obedience-oriented child-rearing values weight hierarchical social roles (affecting Age and Social Value). Crucially, SWA-DPBR does not require exact causal mapping: the 10-dimensional ensemble generates diversity in logit-space gaps, and the IS stage selects the correction balancing collective utility. A rigorous per-dimension causal analysis (e.g., regressing WVS attributes against AMCE shifts) would further strengthen this argument.

C PT–IS, Sampling, and Dual-Pass Reliability

C.1 Utility shape and the omitted explicit KL-on-ϵ term

The scalar decision gap δ acts as a one-dimensional state and the perturbations ϵ_k as control inputs at a single horizon step. Under the path-integral / free-energy view [Williams et al., 2018], softmax weights $w_k \propto \exp(U_{\text{total}}/\eta)$ can be read as targeting a Boltzmann distribution over candidate perturbations. **Our released code does not add** a separate quadratic penalty $\propto \epsilon_k^2/(2\sigma^2)$ to U_{total} in Eq. 7: the released controller matches it exactly, and the Gaussian proposal $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ already limits how far $\tilde{\delta}_k$ can move from $\bar{\delta}$. An optional explicit KL-on-control term remains a straightforward extension if future work needs sharper tail control on $|\epsilon_k|$.

Table 7: SWA-DPBR hyperparameters. Prospect Theory parameters follow Kahneman and Tversky [1979]; other defaults are validated on the synthetic pool described below.

Parameter	Symbol	Value
Persona agents	N	4
IS samples / pass	K_{half}	64 (total $2K_{\text{half}} = 128$)
Reliability scale	s	0.04
Perturbation std. dev.	σ	0.3
Cooperation weight	λ_{coop}	0.7
IS softmax temp.	η	0.5
ESS guard threshold	ρ_{eff}	0.1
PT curvature	$\alpha{=}\beta$	0.88
PT loss aversion	κ	2.25
Decision temp.	T_{dec}	0.5
T_{cat} (Species / Gender / other)	-	4.0 / 3.5 / 1.5
Default logit divisor	T_{logit}	3.0
Country prior	$N_{\text{warm}}, \tau, \beta_{\text{ema}}$	50, 100, 0.1
ESS anchor blend	-	on (off: $\bar{\delta} + \delta^{\star}_{\text{IS}}$ only)

C.2 Comparison with Simpler Alternatives

Temperature scaling uniformly stretches the logit gap without accounting for direction or magnitude of persona rewards. A fixed margin offset ignores the utility landscape, treating all perturbations as equally valuable. PT-weighted IS performs an *adaptive, scenario-specific* correction via nonlinear importance weighting. Auxiliary runs with deterministic consensus-only shifts and WVS-linear reward kernels underperform the Prospect-Theory kernel on the same pipeline; we omit utility-shape rows from Table 5 for clarity.

C.3 Why split K into two passes?

Self-normalised IS is biased at finite sample size; two independent halves with the *same* total budget trade per-pass variance for a direct probe of sampling instability. The reliability map $r = \exp(-(\delta^{\star}_1 - \delta^{\star}_2)^2/s)$ is intentionally smooth-unlike a hard veto it preserves a single code path and down-weights only when the two estimates contradict each other strongly. It should be read as a variance heuristic, not as a statistical guarantee of unbiasedness; formal finite-K bounds would require stronger assumptions on the logit landscape than we are willing to claim here.

C.4 Positional Debiasing: Implementation Detail

The two-pass debiasing of §3.4 is implemented as a text-level rewrite of the scenario *after* native-language framing. To avoid the trivial bug of double-substitution (replacing LEFT $\rightarrow$ RIGHT immediately followed by RIGHT $\rightarrow$ LEFT would leave every label as LEFT), we use a placeholder-based three-step swap, applied independently for the lane-label pair and the group-label pair:

1. Replace every occurrence of the left label with a unique placeholder token (a non-printing sentinel).

2. Replace every occurrence of the right label with the left label.

3. Replace the placeholder with the right label.

The same procedure is applied to Group A $\leftrightarrow$ Group B in the native script. Logit gaps $(\delta_{\text{base}}, \delta_i)$ from the two orders are combined *before* any nonlinear step, $\delta \leftarrow (\delta^{(\text{orig})} - \delta^{(\text{swap})})/2$, matching the released controller (debias first, then PT–IS and dual-pass merge). This guarantees cancellation of additive positional bias in logit space; the final p_{spare} applies $\sigma(\cdot)$ only after that correction.

D Hyperparameter Validation

Implementation. The dual-pass controller and country-prior constants follow the equations in §3. Optional environment overrides can change K_{half}, reliability scale s, and whether the ESS-anchor blend is active; such flags must be applied before controller initialisation when running ablations.

Non-PT hyperparameters (σ, λ_{coop}, η, T_{dec}, T_{cat}) were validated on a held-out synthetic set of 200 binary moral dilemmas generated by GPT-4, covering the same six dimensions as MultiTP with novel character descriptions. Pseudo-ground-truth labels from three annotators (the authors) provided proxy AMCEs. Grid search minimised mean JSD on Qwen2.5-72B. Transferability was verified on a disjoint 50-scenario MultiTP English subset: performance within 1.2% relative JSD of oracle-tuned configuration. T_{cat} values reflect empirical logit magnitude distributions: Species and Gender produce systematically larger gaps, requiring higher temperatures. Because this tuning pool is synthetic and author-annotated, residual bias is possible; we therefore report broad robustness sweeps (Appendix E, Appendix F) to show conclusions are stable beyond one tuned point.

E Temperature Sensitivity

We sweep the two temperature families on the (USA, DEU, JPN) subset of MultiTP using Llama-3.1-70B. Sweep A varies $T_{\text{dec}} \in \{0.10, 0.25, 0.50, 0.75, 1.0, 2.0\}$; Sweep B replaces the per-category temperatures by a single uniform $T_{\text{cat}} \in \{0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 6.0\}$; Sweep C varies the "Others" bucket $T_{\text{cat}} \in \{0.75, 1.0, 1.5, 2.0, 3.0, 4.0\}$ while holding T_{cat}[Species]=4.0, T_{cat}[Gender]=3.5. All three sweeps yield JSD spans below 0.010 (0.0074, 0.0073, 0.0077 respectively), and in every sweep the SWA-DPBR default (†) is strictly *more conservative* than the empirically best value (ΔJSD $\in \{-0.0061, -0.0021, -0.0037\}$). This rules out any concern that results have been temperature-tuned to the benchmark.

Table 8: Temperature sensitivity. Defaults (†) are strictly worse than the best setting in every sweep, providing a conservative lower bound.

Sweep	Setting	JSD ↓	Pearson r ↑	ΔJSD
A: T_{dec}	0.10	0.0502	0.662	+0.0013
	0.25	0.0491	0.658	+0.0002
	0.50†	0.0489	0.630	-
	0.75	0.0476	0.636	−0.0013
	1.00	0.0466	0.641	−0.0023
	2.00 (best)	0.0428	0.627	−0.0061
	JSD span		**0.0074**	
B: Uniform T_{cat}	0.5	0.0521	0.642	+0.0052
	1.0	0.0492	0.673	+0.0023
	1.5†	0.0469	0.700	-
	2.0	0.0461	0.707	−0.0008
	3.0	0.0454	0.707	−0.0015
	4.0	0.0451	0.701	−0.0018
	6.0 (best)	0.0448	0.684	−0.0021
	JSD span		**0.0073**	
C: Others T_{cat}	0.75	0.0526	0.566	+0.0039
	1.00	0.0512	0.589	+0.0025
	1.50†	0.0487	0.634	-
	2.00	0.0475	0.660	−0.0012
	3.00	0.0458	0.693	−0.0029
	4.00 (best)	0.0450	0.706	−0.0037
	JSD span		**0.0077**	

The monotonic JSD reduction with larger T_{dec} reflects the role of T_{dec} in undoing RLHF logit compression: higher values widen the effective token-probability landscape so that IS perturbations exert larger directional influence. However, r peaks near T_{dec}=1.0 and declines at 2.0, indicating a

trade-off between distributional alignment (JSD) and rank-order alignment (r); we keep T_{dec}=0.5 as a deliberately conservative operating point.

F Dataset Preprocessing and Sensitivity

Pipeline. Each MultiTP CSV is processed in five steps: (i) deduplication keeps only the canonical paraphrase (`which_paraphrase = 0`); (ii) the Utilitarianism *quality filter* drops scenarios where the two groups have identical size *and* all characters belong to the *quality-attribute* role set {PREGNANT, WOMAN, LARGEWOMAN}, which would otherwise inject a non-numerosity signal into a dimension whose AMCE is by construction the count-difference effect; (iii) per-category capping retains at most 80 scenarios per dimension; (iv) under-represented categories are oversampled with replacement to a minimum of 36 scenarios; and (v) the resulting pool is shuffled with a fixed seed (42).

Reproducible left/right assignment. When the MultiTP `paraphrase_choice` field does not unambiguously determine which group is rendered on the left vs. right, we deterministically fall back to a SHA-256 hash of the tuple (sub_1, sub_2, g_1, g_2) modulo 2, so the same scenario receives the same ordering across runs and machines. The fallback rate is logged per country; for all 15 countries it stays below 5%.

Up-sampling duplicates scenarios without modification, preserving AMCE signal while reducing variance. Category capping prevents dimension dominance. Side balancing enables effective debiasing. To evaluate whether these choices alter the effective evaluation distribution, we compare *six* preprocessing configurations on 10 representative countries (USA, DEU, CHN, JPN, BRA, VNM, GBR, KOR, RUS, NGA) using Llama-3.1-70B.

Table 9: Per-country dataset sensitivity. JSD range = 0.0034 (aggregate) and < 0.010 in 9/10 countries; the only exception is RUS at 0.013, driven by augmentation sensitivity in the rare Species/Utilitarianism categories. D0 is retained as default.

Country	D0-Default	D1-NoAug	D2-Cap40	D3-Cap120	D4-NoFlip	D5-Strict
USA	0.0655	0.0674	0.0625	0.0644	0.0643	0.0684
DEU	0.0465	0.0515	0.0492	0.0467	0.0444	0.0484
CHN	0.0393	0.0363	0.0385	0.0383	0.0409	0.0398
JPN	0.0304	0.0291	0.0319	0.0296	0.0273	0.0320
BRA	0.0492	0.0551	0.0479	0.0510	0.0520	0.0480
VNM	0.0592	0.0574	0.0631	0.0585	0.0578	0.0626
GBR	0.0614	0.0667	0.0608	0.0618	0.0597	0.0633
KOR	0.0371	0.0354	0.0378	0.0357	0.0349	0.0382
RUS	0.0383	0.0489	0.0389	0.0374	0.0383	0.0357
NGA	0.0522	0.0568	0.0506	0.0525	0.0514	0.0538
Mean	**0.0479**	0.0505	0.0481	0.0476	0.0471	0.0490
Δ vs D0	-	$+0.0025$	$+0.0002$	-0.0003	-0.0008	$+0.0011$

D1-NoAug shows the largest degradation ($+0.0025$), justifying synthetic augmentation of the under-represented Species and Utilitarianism categories. D4-NoFlip is marginally lower than D0, indicating that group-side randomisation is conservative (it adds noise) rather than harmful; we retain it because it preserves positional-debiasing integrity. The 0.0034 aggregate range is well below the ± 0.004 bootstrap CI, so reported results are not confounded by preprocessing choices.

G Cross-Lingual Token Elicitation

Native-language prompting system. Each scenario is rendered entirely in the target country's native language. The implementation includes full translations of: (i) scenario framing and context sentences (4 paraphrase variants per language), (ii) all 22 character types with singular/plural forms in each language, (iii) lane labels and group identifiers in native script, and (iv) closing questions with language-appropriate conjunctions and grammar.

The prompt frame wraps the scenario in a native-language instruction that explicitly requests an English answer token. Each frame follows the pattern: "[moral dilemma preamble in native language]

\n {scenario} \n [instruction to answer A or B in English]". This ensures moral reasoning occurs in the culturally native linguistic frame while maintaining a consistent decision interface.

Token verification. Token IDs are verified per model at initialisation by encoding the uppercase strings `"A"` and `"B"` with `add_special_tokens=False` and asserting single-token mappings: Qwen2.5: A=22397, B=11572; Llama-3.1: A=31266, B=9268. Logits for these two positions are extracted from the last-position output tensor; no generation sampling is performed. The model uses its built-in `chat_template` for correct role formatting across model families.

H Baseline Implementation Details

H.1 Activation Steering

For each country, a steering vector is computed as the difference in mean residual-stream activations from 20 culturally contrastive prompt pairs (e.g., "Answer as someone from [country] who values [WVS-high trait]" vs. "Answer as someone with the opposite values"). Vectors are extracted from layer 32 (transformer midpoint) of Llama-3.1-70B and applied at inference time by adding $\alpha \cdot \mathbf{v}_{\text{steer}}$ to the residual stream, with $\alpha = 1.5$ selected via the same synthetic validation set used for SWA-DPBR.

Sensitivity analysis and fairness considerations. Activation steering performance is known to be sensitive to (i) the extraction layer, (ii) the scaling coefficient α, and (iii) the construction of contrastive pairs [Arditi et al., 2025]. We conducted a limited sensitivity check: extracting from layers 24, 32, and 40 with $\alpha \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$, we found that layer 32 with $\alpha = 1.5$ yielded the lowest mean JSD (0.0877); other configurations ranged from 0.089 to 0.112. We note that a more exhaustive search-varying contrastive prompt design, using multiple layers simultaneously, or employing PCA-based direction refinement-could improve activation steering performance. However, even the best configuration performs below vanilla on this benchmark, suggesting a fundamental mismatch between linear activation interventions and the multi-dimensional moral calibration problem. The key structural advantage of SWA-DPBR is that it operates *per-scenario* with adaptive correction magnitude, whereas activation steering applies a fixed direction uniformly across all scenarios.

H.2 ARGS and controlled decoding (relation to our work)

ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] typically assume trained reward models; a cross-cultural deployment would require a separate reward per target country. On binary forced-choice MultiTP, one can instead define rewards as closed-form functionals of persona logit gaps; a Prospect-Theory kernel aligned with Eq. 7 is then the natural analogue of importance-weighted PT–IS. We do not run the full ARGS decoding stack in our experiments; Table 4 and §6 isolate the kernel comparison on the shared SWA-DPBR pipeline.

I Detailed Ablation Analysis

Without persona: Row 4 removes all WVS-conditioned persona prompts (base-only moral elicitation on the same scenarios). The large JSD lift and Pearson r drop in Table 5 match the expected loss of within-country diversity and numerosity anchoring from the persona ensemble.

Debiasing: Mean positional bias is 0.193 logit units; Species preference inflates from 85.8% to 93.5% without it (human: 79.2%).

J Trigger Mechanism and Latency

SWA-DPBR runs dual-pass IS on every scenario; inter-persona variance is logged but does not gate sampling. The *flip rate* (IS reversals vs. consensus) is 6.0% in the USA ablation (Table 5), indicating confidence modulation rather than wholesale decision reversal.

Mean latency: 730ms (SWA-DPBR) vs. 160ms (vanilla) on Qwen2.5-72B (H100). Overhead from batched multi-agent passes and debiasing; IS operates on cached scalars and contributes negligibly.

K Entropy-Aware Adaptive σ for Qwen2.5-32B

We study whether an entropy-aware adaptive σ can address the Qwen2.5-32B failure mode (Section 5). At each scenario we measure the Shannon entropy H of the top-50 token distribution at the decision position and set $\sigma = \sigma_{\text{base}} \cdot \text{clip}(H/H_{\text{ref}},\ 0.2,\ 5.0)$ with H_{ref}=2.5 nats and σ_{base}=0.30. We compare against a fixed-σ sweep on 8 countries (USA, DEU, CHN, JPN, BRA, VNM, GBR, KOR).

Table 10: Per-country JSD under fixed-σ on Qwen2.5-32B. Two latent country clusters emerge: USA/JPN/GBR/KOR strongly prefer high σ (USA: $0.0523 \rightarrow 0.0317$ at σ=1.0); DEU/CHN/BRA/VNM strictly prefer low σ. All 8 countries share $H \approx 0.52$ nats, so the entropy signal alone cannot discriminate them.

σ	USA	DEU	CHN	JPN	BRA	VNM	GBR	KOR	Mean
0.05	0.0515	0.0647	0.0533	0.0690	0.0692	0.0710	0.0500	0.0670	0.0620
0.10	0.0519	0.0648	0.0532	0.0684	0.0694	0.0711	0.0499	0.0668	0.0619
0.15	0.0522	0.0649	0.0530	0.0676	0.0698	0.0711	0.0499	0.0665	0.0619
0.20	0.0518	0.0651	0.0527	0.0661	0.0702	0.0713	0.0498	0.0661	0.0616
$0.30^{\dagger}$	0.0523	0.0654	0.0516	0.0644	0.0710	0.0717	0.0491	0.0649	0.0613
0.50	0.0487	0.0661	0.0515	0.0601	0.0732	0.0734	0.0476	0.0604	0.0601
0.80	**0.0427**	0.0672	0.0599	**0.0585**	0.0780	0.0757	**0.0439**	**0.0532**	**0.0599**
1.00	<u>0.0317</u>	0.0681	0.0714	0.0637	0.0833	0.0744	0.0458	<u>0.0522</u>	0.0613
Adaptive ($\bar{\sigma}$=0.086)	0.0516	0.0647	0.0532	0.0688	0.0693	0.0711	0.0500	0.0670	0.0620

The mean entropy is $H = 0.521$ nats, far below H_{ref}, confirming concentrated logits. Adaptive σ self-scales to $\bar{\sigma} = 0.086$ ($r(H, \sigma)$=0.948), validating the diagnostic but collapsing onto the low-σ regime. The best fixed σ=0.80 reaches JSD 0.0599 vs. 0.0613 at default-a marginal absolute gain-while the per-country breakdown exposes the deeper problem: countries cluster into a high-σ group (USA/JPN/GBR/KOR) and a low-σ group (DEU/CHN/BRA/VNM), but *all* countries share $H \approx 0.52$ nats, so a per-scenario entropy signal cannot discriminate them. A complete fix therefore requires richer per-country proposal adaptation-left as future work.

L Extended Limitations

Single seed and stochasticity. All main stochastic runs use one seed per model; each scenario runs two independent IS passes of K_{half}=64 proposals (128 total), matching the released pipeline. That dual-pass design provides an internal reliability signal (Eq. 10) but **not** multi-seed confidence intervals for macro MIS; we therefore report headline gains as point estimates and treat seed-level CIs as a priority for future work.

Pearson r in six dimensions. As noted in Section 5, the AMCE vector is short, so per-country r has large sampling error and small row-wise r deltas are not necessarily statistically separable; primary claims use MIS, with JSD as a secondary diagnostic.

WVS-to-trolley linkage. The mapping from WVS value dimensions to trolley moral dimensions follows the indirect-modulation construction in Appendix B.3, not a fully validated causal chain; we therefore avoid claims of the form "WVS dimension X caused AMCE shift Y."

Inter-persona reward assumption. The reward $r_i = \delta_i - \delta_{\text{base}}$ assumes persona shifts approximate human target directions. This is plausible when the base model represents a WEIRD-biased prior, but if a persona's shift is orthogonal to the human target, the signal may mislead. Per-dimension validation is needed. Quantifying whether persona-consensus directions correlate with country-level human AMCE vectors independently of the base model—and when corrections diverge from targets—is also open; we do not report that correlation matrix here.

Geographic coverage, preprocessing, and survey vintage. We report 20 countries in the main MIS sweep (Table 2) out of 100+ MultiTP countries; results depend on the preprocessing pipeline in Appendix F. WVS Wave 7 ($\sim$2017–2022) is a fixed historical slice; persona quality is weakest where coverage is sparse (e.g., Saudi Arabia, Brazil), as reflected in per-country tables. No low-resource languages (Amharic, Khmer, Yoruba) appear in the panel; cross-lingual generalisation to such settings remains untested.

Binary format. The logit-difference formulation requires generalisation to softmax for multi-option settings.

Language–culture conflation. One language per country may conflate linguistic and cultural effects.

Model quantisation. 4-bit/Int8 quantisation may introduce logit artefacts affecting IS stability.

WVS temporal staleness. Wave 7 ($\sim$2017–2022) data may not reflect current preferences in rapidly evolving societies.

Qwen2.5-32B failure mode. Concentrated logit distributions ($H{=}0.521$ nats) cause IS perturbations to land off-manifold and yield near-uniform importance weights. Appendix K shows that an entropy-aware σ correctly diagnoses the cause but cannot fix it on its own, because all 8 evaluated countries share the same low entropy and thus cannot be discriminated by a per-scenario entropy signal. A complete remedy requires richer jointly adaptive σ (and related proposal knobs) per country and is left to future work.

M Per-Dimension Error Analysis

Table 11: Per-dimension worst-error analysis across the 20-country panel. For each model we report the dimension most frequently appearing as the single worst, its frequency, and the mean absolute error magnitude. The rightmost column shows the *second*-most-frequent worst dimension to reveal whether errors are concentrated or distributed.

Model	Worst dim	Freq.	Mean err (pp)	2nd worst dim	Freq.
Llama-3.3-70B	Util_More	15/20	49.7	SocVal_High	5/20
Phi-3.5-mini	Util_More	18/20	42.1	Species_Hum	2/20
Qwen3-VL-8B	SocVal_High	18/20	33.4	Util_More	2/20
Qwen2.5-7B	SocVal_High	12/20	24.5	Species_Hum	5/20
Magistral-24B	SocVal_High	14/20	22.4	Age_Young	3/20
Phi-4	SocVal_High	9/20	23.4	Age_Young	7/20

Table 12: Phi-4 per-country MIS (↓): the best-performing model in absolute terms. SWA-DPBR wins **18/20** countries; the top 10 gains are shown, ordered by relative improvement. The two losses (BRA, IRN) illustrate the diminishing-returns pattern discussed in §7.

Country	Vanilla	DPBR	Gain (%)	Country	Vanilla	DPBR	Gain (%)
USA	0.498	**0.243**	+51.1	JPN	0.395	**0.195**	+50.6
CHN	0.407	**0.214**	+47.4	IDN	0.459	**0.274**	+40.3
GBR	0.503	**0.319**	+36.6	THA	0.456	**0.313**	+31.4
MYS	0.459	**0.328**	+28.5	MMR	0.482	**0.357**	+26.0
BGD	0.489	**0.368**	+24.7	ROU	0.504	**0.385**	+23.5
Losses: BRA $0.274 \rightarrow 0.299$ (-9%)				IRN $0.397 \rightarrow 0.530$ (-34%)			
Macro mean: $0.454 \rightarrow$ **0.346**				+23.7%, 18/20 wins, mean $r = +0.56$			

Three structural insights emerge from these tables.

First, **the bottleneck dimension shifts with model quality**. Weaker models (Llama-3.3-70B, Phi-3.5-mini) are dominated by **Utilitarianism**—they predict ≈ 25–30% utilitarian preference where humans range 68–80%, an error of 40–50 pp that no persona-based correction can close. Better-calibrated models (Phi-4, Magistral, Qwen2.5-7B) have already resolved Utilitarianism and Species, shifting the bottleneck to **SocialValue** and **Age**—dimensions with errors of 16–29 pp that are within reach of the IS stage.

Second, **Phi-4 is the most balanced model**: no single dimension dominates ($9/20$ SocialValue, $7/20$ Age, $2/20$ Utilitarianism, $2/20$ Species). This balance—not raw scale—explains why it achieves the lowest absolute MIS despite being $5\times$ smaller than Llama-3.3-70B.

Third, the per-country Phi-4 table (Table 12) confirms that **gains are large and geographically broad**: 51% on USA and Japan, 47% on China, 40% on Indonesia—with the two losses (Brazil, Iran) tracing to already-low vanilla MIS where IS overshoots.

N Broader Model Landscape

The six models in Table 2 were selected from a broader sweep of **28 model–method combinations** covering 12 distinct architectures. Table 13 summarises the full landscape, ordered by macro improvement versus vanilla.

Table 13: Broader model sweep (5-country prototyping panel: BRA, CHN, DEU, JPN, USA). Models above the line show positive macro gains; those below show degradation. Bolded rows appear in the main paper's 20-country evaluation.

Model	Params	Mean MIS ↓	vs. van. (%)	Win/5
Phi-4	14B	**0.246**	+34.4	4/5
Llama-3.3-70B (4-bit)	70B	0.524	+25.9	5/5
Qwen3-VL-8B	8B	0.381	+23.7	4/5
Llama-3.1-8B (4-bit)	8B	0.455	+16.9	5/5
Magistral-Small-2509	24B	0.317	+13.6	4/5
Gemma-4-E2B	2B	0.426	+11.6	4/5
Qwen2.5-7B (bf16)	7B	0.385	+8.2	3/5
Phi-3.5-mini	3.8B	0.545	+7.3	4/5
Mistral-7B-v0.3	7B	0.428	+5.9	5/5
Qwen3.5-0.8B	0.8B	0.471	+3.2	4/5
Gemma-7B	7B	0.432	+2.5	3/5
Gemma-3-270M	270M	0.462	+1.3	4/5
Llama-3.2-1B	1B	0.476	+1.1	1/5
9 additional models showed ≤ 0 macro gain (omitted; see §7)				

Several patterns inform the main paper's model selection. *Success correlates with logit well-behavedness, not raw scale:* Phi-4 (14B) leads while Gemma-4-31B (31B) and Qwen3-Coder-30B (30B) degrade—confirming that SWA-DPBR requires a cooperative decision surface, not just more parameters. *Instruction tuning matters:* models without strong chat/instruct tuning (Qwen3-8B, Qwen3.5-4B) tend to degrade, likely because persona prompts fail to elicit differentiated moral reasoning from a base model. *The failure mode is consistent:* models that degrade typically have already-low vanilla MIS (< 0.40), leaving insufficient headroom for IS corrections—matching the "diminishing returns" pattern discussed in §7.

O Engineering Journey: Method Selection

SWA-DPBR (the "EXP-24" configuration) was selected from **25 method variants** developed and evaluated on a fixed three-model, five-country prototyping panel (Qwen2.5-7B, Gemma-2-9B, Mistral-7B $\times$ USA, CHN, JPN, DEU, BRA). Table 14 summarises the top-performing variants.

Table 14: Method variant comparison on the 3-model $\times$ 5-country prototyping panel (15 rows). Mean MIS ↓. All variants share the same persona construction and debiasing; they differ in the IS/prior stage.

Rank	Method variant	Mean MIS ↓	Key innovation
1	EXP-23: Category-Coherence Reg.	0.3956	Blend IS toward per-cat mean
2	**EXP-24: Dual-Pass Bootstrap (DPBR)**	**0.3969**	Soft $r = \exp(-v_b/s)$ reliability
3	EXP-09: Hierarchical IS	0.3975	Country EMA prior
4	EXP-22: Adaptive IS σ	0.3977	σ from agent/history stats
5	EXP-16: Nesterov IS	0.3977	Momentum + progressive PT
6	EXP-17: Dual-Momentum Prior	0.3979	Fast/slow β EMA blend
7	EXP-10: Grand Fusion	0.3982	EXP-03+05+09 combined
18	EXP-01: Base SWA-PTIS	0.4269	Single-pass IS (no prior)
25	EXP-12: Contrastive Personas	0.4517	World-average subtraction

The top cluster (ranks 1–7) is separated by only 0.0026 MIS, but DPBR was selected over the marginally-better EXP-23 for two reasons: (i) DPBR materially **lowers flip rate** (the fraction of

scenarios where IS reverses the consensus decision) compared to all other top-cluster methods, improving deployment stability; and (ii) the dual-pass bootstrap provides an **interpretable uncertainty signal** (r in Eq. 10) that can be logged and audited per scenario. The base SWA-PTIS (rank 18) shows the value added by the hierarchical prior and dual-pass stages: a 7% relative MIS improvement. Contrastive persona decoding (rank 25) *hurts* alignment, validating the design choice discussed in §2 to use direct WVS personas rather than contrastive amplification.